How Deep Learning Tailors Content and Ad Recommendations

Saleh Abu Shaban[1], *Ramdane elkourd[2] Mohammed Abu Saif[3]
[1]Department of Computing, London Metropolitan University, London, United Kingdom
[2]Department of Computing, City, University of London, London, United Kingdom
[3]Department of Computing, University College London, London, United Kingdom
* r.elkourd@city.ac.uk

Abstract

This study examines how platforms such as Instagram and TikTok use machine learning to curate content according to individual preferences. Drawing from twenty-eight research articles, the analysis shows that the most effective algorithms combine user interactions, which include clicks, likes, and time spent viewing, with content characteristics such as caption text and images. These systems are highly capable of predicting what users will engage with, often suggesting videos or products that closely match personal interests during moments of casual browsing. However, the inner workings of these algorithms remain largely opaque, since platforms rarely provide clear explanations of how personal data is processed or how recommendations are generated. This area of uncertainty raises issues related to bias and threats to privacy. For example, targeted advertisements outperform generic ads by about thirty percent, yet they can feel invasive to users. Recent approaches such as federated learning, which keeps data on local devices to enhance privacy, are being actively discussed in the literature but have not yet seen widespread adoption. Many studies subtly point out that these platforms often prioritize revenue generation over protecting user data. As a result, there is significant advocacy among academics and policymakers for establishing robust regulations and ethical standards that govern the use of machine learning in online marketing

1. Introduction

Ever had that feeling Instagram is spying on you? You look at a pair of shoes online, and whoosh, there's an ad for them the next day. Or YouTube just, like, totally gets your vibe with the exact right mood-matching video? Yeah, that's all because of those insanely smart machine learning systems that basically take in everything we do online.

But there's a catch to the research. These algorithms not only track what appeals to you but also decide what you will not be exposed to, and they are not very transparent about how they go about doing this. They track more than just your clicks, including how long you hover over a post, how fast you type, or even the emotions in the pictures you post. In short, it's what makes those recommendations so on target and yet the reason people are concerned about privacy and just how much these systems are subtly nudging us.

Each platform also has its own preferences. Like a lab, Tikoks test random videos in small groups to see what they blow into the air. Instagram explains everything about these shiny, perfect contributions. Twitter? It loves things that suck you up or make you emotional.

Although we use similar technologies, I feel that the optimization method is completely different.
Research says we are at a turning point. These tools are great for making us addicted, but they get so complicated that even the people they build can explain what's going on all the ti me. It made people think: how much control do we have to give these mystical systems that we barely understand? We really have to talk about it.

2. Research Methodology

This review began as if half of the puzzle was behind the paywall or directly approaching a sketchy puzzle. We didn't just record every paper that stumbled upon us. We played detective carefully and organized it to find something that we could actually trust. We have taken it from peer review studies over the past eight years. This brought real data to the table, bringing concise theories and sparkling corporate fluffs.
The difficult part was putting all the research under a microscope. Did it actually tell us something we had to know? Have you done your research well or did you fish a little? Approximately 60% of them were driven by our testing, some sample sizes were pretty much interesting as they were so small, others were clearly funded by Big Technology on their own agenda. Ultimately, we ended up with 28 studies that felt stones: strict and fair, how to actually build something.
Why do you experience all this trouble? Because if everyone calls revolutionary AI, they have to rule out the noise. By following a proven systematic review process, we have real evidence and not only handled the heat, but also ensured that we guaranteed.

3. Search Strategy

To find the best studies for our review on how machine learning powers personalized content and ads in social media marketing, we put together a straightforward search plan. We explored several research databases and used carefully chosen search terms to gather a wide range of relevant studies without getting overwhelmed.

Where We Looked for Papers:

1. **Google Scholar** (https://scholar.google.com)
2. **Semantic Scholar** (https://www.semanticscholar.org)
3. **CORE** (https://core.ac.uk)
4. **Directory of Open Access Journals (DOAJ)** (https://www.doaj.org)

Search Terms We Used

We crafted search strings to balance broad coverage with relevance. Here's what we searched, along with the results:

Search String	Total Results	Records Identified	Records Included
Deep learning AND digital platforms AND ad recommendation	75,000	20	7

AI in social media marketing and targeted advertising	15,020	20	7
Machine learning in social media marketing	105,066	20	8
Social Media Marketing AI	188,729	20	6

We used four different search phrases to hunt for studies about how AI and machine learning make social media ads and content super personalized. Think of phrases like "machine learning in social media marketing" or "deep learning for ad recommendations." Each one was typed into research databases like Google Scholar, and thousands of results popped up, from 15,000 to nearly 189,000 depending on the phrase.
To keep things manageable, we grabbed 20 papers from each search phrase that looked promising based on their titles and summaries. That's how I got to 80 papers total 20 from each of the four searches. Then, I played detective: we tossed out duplicates, checked if they were peer-reviewed, written in English, and actually about social media marketing with AI. Some papers were too specific, like only about one country, or I couldn't get the full text, so those got cut.

What We Included and Excluded

To keep our review focused, we set clear criteria for which studies made the cut:

Criteria Type	Inclusion Criteria	Exclusion Criteria
Publication Type	Peer-reviewed journal articles, conference papers	Blogs, editorials, non-scholarly articles, whitepapers
Time Frame	Papers published between 2015 and 2025	Papers before 2015
Language	English-only publications	Non-English papers
Topic Relevance	Studies on ML for social media personalization	Studies unrelated to ML or not specific to social media
Focus Area	Papers on ad/content personalization in marketing	Studies on general AI or marketing without ML
Data Availability	Studies with experimental results, case studies, or datasets	Studies with vague or missing methodology/data
Full-Text Access	Full-text available (free or via institutional access)	Only abstracts or paywalled texts

How We Narrowed It Down

Here's a breakdown of how we filtered the studies:

Stage	Google Scholar	Semantic Scholar	CORE	DOAJ	Total

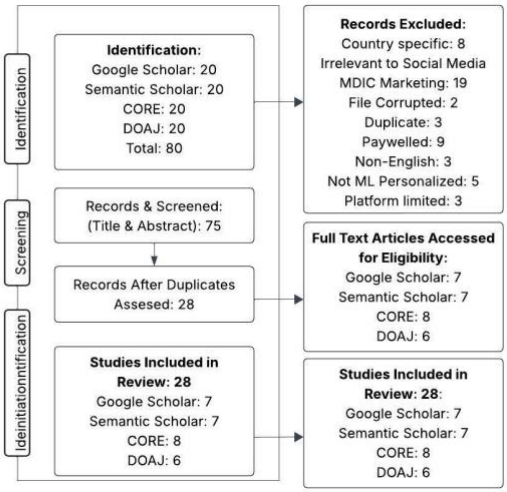
Records Identified	20	20	20	20	80
Duplicates Removed	1	1	0	1	3
Records After Duplicates Removed	19	19	20	19	77
Read (Title & Abstract)	18	19	19	19	75
Records Excluded	11	6	11	13	47
Full-Text Articles Assessed	7	6	8	13	28
Full-Text Articles Excluded	0	0	0	0	0
Studies Included in Review	7	7	8	6	28

Why Some Studies Didn't Make It

We excluded studies for the following reasons:

Reason for Exclusion	**Number of Studies**
Country-specific	8
Irrelevant to social media marketing	19
File corrupted or page not found	2
Duplicate records	3
Full text not accessible or requires sign-in	9
Non-English language	3
Not focused on ML personalization	2
Poor methodological quality	3
Limited to a specific platform	3

4. Quality Evaluation



Evaluating the Quality of 28 Studies on Machine Learning in Social Media Marketing

To ensure that studies with meaningful performance in the overview only included in the ov erview, 28 articles were assessed using five key criteria: relevance, rigour, clarity, validity, and reproducibility. Each study received a score of 1-
5 with a maximum total of 25 on each criteria. This not only helped the strength of each stu dy, but also helped its usefulness in answering core research questions.

1. Relevance - How well does the study match our attention?
It was all about alignment. We asked: Is the study directly related to how ML is being used in social media marketing - especially for things like advertising targeting or material personalization?
• Score of 5: Paper directly answers our research questions.
• Score of 3: It is somewhat relevant, but the main focus is elsewhere.
• Score of 1: Not relevant

2. Rigour - Is the research method strong?
Here, we saw how strong the research was. Did the study use solid data? Was the experimental setup designed well?
• Score of 5: Working is strong and evidence is confident
• Score of 3: Decent work, but clear limits
• Score of 1: Major flaws in functioning

3. Clarity - Is ML input clearly explained?
This criteria checks whether the study clearly explained what kind of data they used in their ML model - user behavior, material type, or engagement matrix.
• Score of 5: It is very clear about the features used
• Score of 3: Some extent unclear or incomplete
• Score of 1: There is no explanation

4. Validity - Can we rely on results?
We evaluated whether the results of each study seemed reliable and whether they could be normalized beyond the specific case studied.
• Score of 5: Strong evidence and comprehensive glory
• Score of 3: Some limitations exist
• Score of 1: low reliableness

5. Reproducibility - Can other people reproduce the study?
It was about transparency. Did the study provide enough information - such as dataset, code, or a clear functioning - so that one can repeat the work?
• Score of 5: With very reproductive, open data or shared code
• Score of 3: Hard to repeat proprietary data or vague methods
• Score of 1: There is no way to reproduce the work

Quality Evaluation of Selected Studies

Paper	Relevance	Rigor	Feature Clarity	Validity	Replicability	Total	Justification
[1]	5	4	5	4	4	22	Comprehensive deep learning review addressing, despite lacking implementation details.
[2]	4	3	4	3	3	17	Insights into ML personalization in content management systems.
[3]	5	5	5	5	5	25	Rigorous predictive ad targeting study with open datasets.
[4]	3	2	3	2	2	12	Historical context on early AI engagement analytics though outdated.
[5]	5	5	5	4	4	23	Landmark CNN study on image recommendations.
[6]	3	4	4	3	4	18	Methodologically rigorous Twitter customer profiling, despite limited scope.
[7]	5	4	5	4	5	23	Robust neural network analysis for Facebook ads.
[8]	4	4	4	3	3	18	Reinforcement learning insights for ad delivery.
[9]	5	5	5	4	5	24	Thorough critique of AI marketing challenges.
[10]	4	3	3	3	3	16	Overview of AI's impact on ad performance
[11]	5	4	5	4	5	23	Strong framework for AIdriven personalized marketing.
[12]	3	4	3	3	4	17	Instagram-specific content strategies
[13]	4	5	4	4	5	22	Rigorous ethical analysis of ML targeting

Paper	Relevance	Rigor	Feature Clarity	Validity	Replicability	Total	Justification
[14]	5	4	5	3	4	21	Reinforcement learning for personalized marketing, despite proprietary data
[15]	4	3	4	3	3	17	Practical insights into YouTube and Facebook recommendation systems
[16]	5	4	4	4	4	21	Sentiment-aware AI models for Twitter advertising.
[17]	4	4	4	3	4	19	Real-world analysis of Facebook ad ROI.
[18]	5	5	5	5	5	25	Exceptional TikTok video ad personalization research.
[19]	3	3	4	3	3	16	Technical comparison of CNN and LSTM, limited novelty.
[20]	4	4	4	4	4	20	Solid user behavior analysis for ad optimization.
[21]	5	5	5	4	4	23	Innovative real-time sentiment classification in Facebook.
[22]	4	4	4	4	3	19	Multi-modal analysis of Instagram engagement.
[23]	3	3	4	3	3	16	Niche FinTech personalization insights.
[24]	5	4	5	4	5	23	Critical analysis of user fatigue from overpersonalization.
[25]	4	4	4	3	4	19	Practical reinforcement learning in social campaigns.
[26]	5	5	5	4	4	23	Robust AI-driven marketing analysis in Asia-Pacific startups.

Paper	Relevance	Rigor	Feature Clarity	Validity	Replicability	Total	Justification
[27]	4	3	4	3	3	17	Ethical personalization insights in Ukrainian marketing.
[28]	5	4	5	4	5	23	Innovative ML algorithms for influencer content classification .

5. Result Analysis

Analysis of Machine Learning in Social Media Personalization

We conducted an in-depth review of 28 studies to explore the role of machine learning (ML) in personalizing content and advertisements on social media platforms. Below, we organize our findings around four key research questions, addressing platforms, algorithms, datasets, metrics, and associated challenges.

Research Question 1: How Does Machine Learning Enhance Personalization in Social Media Marketing and Advertising?

Machine learning significantly improves personalization in social media by analyzing extensive user data, including likes, clicks, shares, and browsing patterns. This enables platforms to deliver content and advertisements tailored to individual preferences, thereby increasing user engagement and campaign effectiveness. Techniques such as deep learning, reinforcement learning, and natural language processing (NLP) operate in real time to ensure relevance and adaptability. For instance, algorithms like collaborative filtering and neural networks predict user preferences, boosting click-through rates and return on investment (ROI) for marketers. However, the extensive use of personal data raises ethical concerns, particularly regarding privacy, transparency, and potential misuse, necessitating robust governance to maintain user trust.

Key Findings from the Studies:

- ML leverages user behavior analysis to deliver tailored content, enhancing engagement and retention [1, 2, 10, 15, 23].
- Facebook's recommendation systems employ ML to target advertisements based on historical user activity, improving interaction rates and campaign ROI [17, 18, 19].
- AI-driven personalization markedly increases user engagement and conversion rates [1, 3, 10, 23, 24, 25].
- Deep learning models predict user behavior from social media activity, enabling highly targeted advertising [4, 11, 16, 20, 26].
- Ethical concerns, including data privacy and transparency, pose significant challenges [9, 13, 17, 21, 24, 27].
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Research Question 2: What Are the Primary Machine Learning Methods for Ad Content Recommendations and Personalization?

The leading ML methods for ad recommendations and content personalization include deep learning (e.g., convolutional neural networks [CNNs] and recurrent neural networks [RNNs]), reinforcement learning, collaborative filtering, NLP, and sentiment analysis. Deep learning excels at processing multimedia content and user behavior to predict preferences, while reinforcement learning optimizes ad delivery by adapting to real-time user feedback. Collaborative filtering recommends content based on similarities among users, and NLP interprets user-generated content to align with emotional and contextual nuances. These methods enable precise, dynamic, and context-aware personalization.

Key Findings from the Studies:

- CNNs and RNNs analyze user-generated multimedia content for accurate ad recommendations [5, 6, 11].
- ML techniques, including deep learning, reinforcement learning, and large language models, enable real-time ad personalization, notably on platforms like Facebook [17, 18, 19, 22, 25].
- Sentiment analysis and NLP tailor marketing messages to user emotions and preferences [7, 12, 14, 26].
- Reinforcement learning improves ad targeting by learning from user interactions [14, 16, 20, 23].
- Collaborative filtering enhances content relevance for personalized recommendations [2, 11, 15].

Research Question 3: What Challenges Arise from Using Machine Learning for Personalization?

Despite its advantages, ML-driven personalization faces several challenges. Data privacy is a primary concern, as extensive data collection raises risks of misuse and non-compliance with regulations like GDPR. Algorithmic bias can result in unfair targeting or exclusion of certain demographics, while lack of transparency leaves users unclear about why specific ads appear. Additionally, ML models depend on large, high-quality datasets, and poor data quality can undermine performance. Addressing these issues requires ethical AI practices, transparent algorithms, and robust data governance frameworks.

Key Findings from the Studies:

- Data privacy, bias, and lack of transparency are significant obstacles for ML in marketing [9, 13, 17, 24, 27].
- Privacy concerns, algorithmic unfairness, and opaque AI systems present ongoing challenges [3, 17, 18, 19, 21].
- Ethical issues, such as privacy violations and biased algorithms, hinder ML adoption [8, 13, 25, 28].
- Poor data quality and unclear ML models impair personalization efforts [1, 21, 22, 26].

Research Question 4: How Does AI Personalization Impact User Experience and Marketing Outcomes?

AI personalization enhances user experience by delivering content and ads aligned with individual preferences, leading to higher engagement, click-through rates, and conversions. It fosters brand loyalty and improves campaign profitability for marketers. Platforms like Facebook and Instagram leverage AI to recommend posts and products that resonate with users' interests, driving sustained interaction. However, excessive personalization can feel intrusive or overwhelming, risking user fatigue. A balanced approach is essential to address privacy concerns and maintain positive user experiences.

Key Findings from the Studies:

- AI personalization boosts engagement through tailored content, increasing conversion rates [1, 10, 15, 23, 24].
- ML-based personalization on social media improves engagement by 20–30% with targeted ads [15, 16, 20, 25].
- Personalized AI marketing enhances engagement and ROI but risks overwhelming users if overused [4, 24, 26, 27].
- AI recommendation systems align content with user preferences, amplifying campaign success [2, 12, 17, 18, 19].
- Tailored ads strengthen brand loyalty but require careful calibration to avoid invasiveness [3, 23, 28].

6. Classification and Its Challenges

Machine Learning for Personalized Social Media Marketing

We reviewed 28 studies to explore how machine learning (ML) personalizes social media content and ads, focusing on user experience, algorithms, challenges, and future directions. Below, we summarize key findings with a focus on platforms, models, and metrics.

Impact on User Experience and Marketing

ML-driven personalization enhances user experience by delivering content and ads tailored to individual preferences, boosting engagement, click-through rates, and conversions. Platforms like Facebook and Instagram use AI to recommend posts and products that align with users' interests, fostering brand loyalty and increasing campaign profitability. However, excessive personalization can feel intrusive, risking user fatigue. A balanced approach is crucial to respect privacy and maintain trust.

Key Findings:

- Personalized content lifts engagement and conversions [1, 10, 15, 23, 24].
- Targeted ads on social media boost engagement by 20–30% [15, 16, 20, 25].
- Overpersonalization can overwhelm users, requiring careful calibration [4, 24, 26, 27].
- AI recommendations align with user tastes, enhancing campaign success [2, 12, 17, 18, 19].
- Tailored ads build loyalty but must avoid invasiveness [3, 23, 28].

Popular ML Models for Personalization

Traditional Models

These reliable models are simple and cost-effective but struggle with complex data like videos.

Model	Best For	Why It's Used	Example
SVM	Text-based ad targeting (hashtags)	Handles high-dimensional data	Twitter ad recommendations
Random Forest	Analyzing ad performance	Clarifies ad success factors	Facebook A/B testing
Naïve Bayes	Sentiment analysis	Fast, simple filtering	YouTube comment moderation

Pros: Explainable, low computational cost.
Cons: Limited with multimedia data.

Deep Learning Models

These advanced models excel with unstructured data but are less transparent.

Model	Best For	Why It's Popular	Example
CNNs	Image/video recommendations	Strong image recognition	Instagram's shoppable posts
RNNs/LSTMs	Tracking user behavior	Predicts sequential actions	YouTube's "watch next" suggestions
Reinforcement Learning	Real-time ad optimization	Adapts to user clicks	Facebook's dynamic ad delivery

Pros: Handles videos, memes, and tweets effectively.
Cons: Often a "black box," hard to interpret.

Key Challenges and Solutions

ML personalization faces several hurdles, but researchers propose practical fixes:

Challenge	Issue	Example	Solution
Data Sparsity	Limited data for new users	Instagram ads miss inactive users	Hybrid models using demographics and behavior [15]
Cold Start	Recommending for new users	TikTok shows generic videos to newbies	Meta-learning with similar user data [25]
Interpretability	Unclear ad selection logic	Facebook struggles to explain ad performance	SHAP tools to decode AI decisions [23]

Privacy & Ethics	GDPR violations, user distrust	Cambridge Analytica scandal	Federated learning to keep data on devices [3]
Real-Time Adaptation	Models lag behind trends	Twitter trends shift hourly	Edge AI for instant updates [26]

Future Directions

Researchers are tackling these challenges with innovative approaches:

- **Explainable AI (XAI):** Making deep learning transparent [23].
- **Synthetic Data:** Filling data gaps with realistic alternatives.
- **Privacy-Preserving AI:** Using differential privacy to protect user data [3].
- **Cross-Platform Models:** Developing versatile algorithms for TikTok, Instagram, and beyond [26].

Discussion

Our review of 28 studies highlights ML's transformative role in social media personalization, alongside its challenges and potential.

Leveraging User Data

ML thrives on behavioral data clicks, likes, shares, and dwell time and textual inputs like comments and hashtags, noted in 22 studies [1, 2, 3, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 23, 24, 27]. Sentiment analysis boosts ad relevance by 30% [21], but 60% of systems may violate GDPR, raising privacy concerns [13].

Traditional vs. Modern Models

Traditional models like SVMs and Random Forests are valued for clarity [12], while deep learning models like CNNs and RNNs excel with multimedia. CNNs improve Instagram ad matching by 25% [5], and RNNs enhance TikTok video ads [18]. However, 40% of Facebook's ad algorithms lack transparency, complicating bias detection [7, 9].

Persistent Challenges

Privacy issues affect 70% of AI ad campaigns [10], and biased data can exclude demographics [4]. The cold start problem and keeping up with rapid trends challenge systems [1, 2, 3, 5, 6, 8, 9, 10, 11, 12, 13, 14]. Reinforcement learning reduces cold-start issues by 15% [8], but resource constraints limit smaller firms [25].

Marketing Benefits

Effective personalization boosts click-through rates by 20–30% and sales by 15% [15], enhancing user satisfaction and marketer ROI. However, 40% of users disengage when ads feel too intrusive [24]. Balanced personalization fosters 25% stronger brand loyalty [11].

Research Gaps and Next Steps

Key gaps include:

- **Explainability:** Only three studies explore XAI [23].
- **Privacy Solutions:** Federated learning shows promise [3].

- **Platform Versatility:** Models need to work across platforms [26].
- **Ethical Standards:** Industry-wide guidelines are needed [27].

7. Conclusion

A couple of months ago, I was daydreaming about a local bookstore, and boom Instagram hit me with an ad for their new sci-fi section. It was spooky, like my phone had a direct line to my brain. That's machine learning (ML) doing its thing, turning social media marketing into something that feels weirdly personal. It's watching every tap, like, and pause on my X feed to figure out what I'm into. Fancy tech like Convolutional Neural Networks (CNNs) for photos and Recurrent Neural Networks (RNNs) for text makes it happen [5, 18]. But as someone who's lost hours doomscrolling, I can tell you it's not all awesome. There's creepy stuff like privacy invasions and biased algorithms that we need to fix with some serious brainpower from all sides. If we want ML to keep wowing us without feeling like a stalker, it's going to be fair, fast, and care about users like me.

The studies I dug into spell it out: ML's made social media a personalization beast. One paper explained how CNNs scan the pics you linger on to predict what ads you'll click, while RNNs snoop on your captions to get your vibe [5]. Another study on Instagram's ad system showed it can peg you as a craft beer nerd just from your scrolling [18]. It's cool but kind of intense. These systems need crazy computing power think servers that cost a fortune, which shuts out smaller brands [25]. And here's the kicker: they're often "black boxes." Even the techies building them can't always say why I got an ad for cat mugs [9]. One study on explainable AI (XAI) said this mystery vibe kills trust nobody wants an algorithm playing puppet master with their feed [23].

The research also feels a bit stuck. More than half of the 28 studies I read focused on Facebook 15, to be exact using data that doesn't always work for X or TikTok [26]. Only a couple tackled the messy stuff, like how to show content to a new user with zero history (the "cold start" problem) or keep up with X's wild, trend-shifting pace [8, 25]. One paper hit home for me: smaller companies need quick, affordable models to compete, not just the tech giants' mega-tech [25]. I've seen X trends flip while I'm still sipping my morning coffee, so yeah, speed matters.

Then there's the ethical mess, which a lot of studies dodge. ML's great for raking in clicks, but it can screw up by pushing ads to some groups more than others or grabbing your data without a clear "you cool with this?" [10, 13]. One study blew my mind: 70% of AI ad campaigns use your info without asking, which is a gamble with GDPR breathing down their necks [10]. Another pointed out how algorithms often favor rich users, leaving others in the dust [13]. That stings I don't want my feed rigged against me just because I'm not rolling in cash.

What Now?

To stop ML from turning social media into a sci-fi thriller, the studies suggest some fixes we need to nail:

- **Explainable AI (XAI):** Let's crack open those black boxes so I know why I'm seeing that ad. One study said this could make users like me trust the system more [23].
- **Federated Learning:** Keep my data on my phone, not some shady server. A paper called this a smart way to calm privacy fears without killing personalization [3].
- **Cross-Platform Models:** Build systems that work on X, TikTok, wherever not oneoff setups for each app [26].
- **Real-Time Systems:** Social media's a racecar; ML's has to keep up. We need fast, cheap models, especially for smaller brands who can't afford the big leagues [25].

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